

[Bugfix] Migrate to REGEX Library to prevent catastrophic backtracking #18454

New issue

Merged

simon-mo merged 10 commits into `vllm-project:main` from `Crucifixion-Fxl:main` on May 24, 2025

Conversation17

Commits10

Checks4

Files changed52

+62-58



Crucifixion-Fxl commented on May 21, 2025

Contributor

• edited by github-actions (bot)

Replace regex library to improve parsing performance

FIX [#18304](#): [Bug]: tool parsers can block the async event loop

- re library

```
~ time python test_re.py
match
python test_re.py 46.02s user 0.00s system 99% cpu 46.056 total
```

\n

- regex library

```
~ time python test_re.py
match
python test_re.py 0.04s user 0.01s system 98% cpu 0.049 total
```

If anything seems incorrect, please point it out and I'll fix it right away.

2

Reviewers

	aarnphm	
	mgoin	
	russellb	
	DarkLight1337	
	ywang96	
	robertgshaw2-redhat	
	simon-mo	
	jeejeelee	
	tlrmchlsmth	
	WoosukKwon	
	njhill	
	comaniac	
	alexm-redhat	
+1 more reviewer		
	dtrifiro	



Crucifixion-Fxl requested review from DarkLight1337, jeejeelee, mgoin, robertgshaw2-redhat, russellb, simon-mo, tlrmchlsmth and ywang96 as code owners 8 months ago

Assignees

No one assigned

Labels

- ci/builddocumentationfrontend
- multi-modalityreadystructured-output
- tool-callingv1

Projects



github-actions (bot) commented on May 21, 2025

Hi! Thank you for contributing to the vLLM project.

Join our developer Slack at <https://slack.vllm.ai> to discuss your PR in #pr-reviews, coordinate on features in #feat-channels, or join special interest groups in #sig- channels.

Just a reminder: PRs would not trigger full CI run by default. Instead, it would only run `fastcheck` CI which starts running only a small and essential subset of CI tests to quickly catch errors. You can run other CI tests on top of those by going to your `fastcheck` build on Buildkite UI (linked in the PR checks section) and unblock them. If you do not have permission to unblock, ping `simon-mo` or `khluu` to add you in our Buildkite org.

Once the PR is approved and ready to go, your PR reviewer(s) can run CI to test the changes comprehensively before merging.

To run CI, PR reviewers can either: Add `ready` label to the PR or enable auto-merge.

□

 **Structured Output** 

Status: Done +6 more

 **Tool Calling** 

Status: Done +6 more

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

✓ **[Bug]: tool parsers can block the asyn...**

7 participants



  **mergify** (bot) added `documentation` `ci/build` `frontend` `multi-modality` `structured-output` `v1` labels on May 21, 2025

  **github-project-automation** (bot) added this to **Structured Output** on May 21, 2025

  **mergify** (bot) added the `tool-calling` label on May 21, 2025

  **github-project-automation** (bot) added this to **Tool Calling** on May 21, 2025

  **Crucifixion-Fxl** force-pushed the `main` branch from `52de3ee` to `1a78a2c` 8 months ago Compare



 **mgoin** approved these changes on May 22, 2025 View reviewed changes

mgoin left a comment

Member

LGTM - `regex` seems like a solid library and is even recommended in the [docs for re](#)

See also The third-party [regex](#) module, which has an API compatible with the standard library [re](#) module, but offers additional functionality and a more thorough Unicode support.

It has many precompiled wheels available and no dependencies: <https://pypi.org/project/regex/#files>

cc [@aarnphm](#) [@russellb](#) for visibility, but I think this is a pretty clear win

 1



mgoin requested a review from **aarnphm** [8 months ago](#)



mgoin requested changes
on [May 22, 2025](#)

[View reviewed changes](#)

mgoin left a comment

Member

@Crucifixion-Fxl it seems like the api isn't fully compatible since this test failure is related
<https://buildkite.com/vllm/fastcheck/builds/24585/steps?jid=0196f270-992a-4385-b629-aba22b50ad20#0196f270-992a-4385-b629-aba22b50ad20/9841-14182>



mgoin commented on [May 22, 2025](#)

Member

Also you could just replace the import rather than replacing all the usage, as in "import regex as re"



Crucifixion-Fxl commented
on [May 22, 2025](#)

Contributor

Author

Also you could just replace the import rather than replacing all the usage, as in "import regex as re"

Thank you for your suggestion. I will try to revise it according to your advice.

 1



[\[Bugfix\] Migrate to REGEX Library to prevent catastrophic backtracking](#) ...

 [2d9ca06](#)



Crucifixion-Fxl [force-pushed](#) the [main](#) branch from [f4b13c2](#) to [2d9ca06](#) [8 months ago](#)

[Compare](#)



aarnphm requested changes
on May 22, 2025

[View reviewed changes](#)

aarnphm left a comment

Collaborator

Please fix one comments about the structured outputs script,
otherwise lgtm

benchmarks/benchmark_serving_structured_output.py

Outdated

	690	+	import <u>regex</u> as re
691	691		
692		-	return re.match(args. <u>regex</u> ,
	692	+	return re.match(args. <u>re</u> , act



aarnphm on May 22, 2025

Collaborator

This has to still be regex



Crucifixion-Fxl

on May 22, 2025

Contributor

Author

Thank you for the reminder. I've already fixed it



aarnphm commented on May 22, 2025

Collaborator

Can we enforce Version 0 for now to make sure that it
compatibles with re?

seems like there are some different with version 1 here
<https://github.com/mrabarnett/mrab-regex#old-vs-new-behaviour>



[\[Bugfix\] Migrate to REGEX Library to prevent catastrophic backtrackin...](#) 8de207a



Crucifixion-Fxl commented

Contributor

Author

on May 22, 2025

Can we enforce Version 0 for now to make sure that it
compatibles with re?

seems like there are some different with version 1 here
<https://github.com/mrabarnett/mrab-regex#old-vs-new-behaviour>

This seems acceptable at first glance, but I'll review it carefully.

11 hidden items

[Load more...](#)

 Merge remote-tracking branch
'upstream/main' ✖ [7bfb16a](#)

 **mergify** bot removed the **needs-rebase** label
on May 23, 2025

 **Crucifixion-Fxl** added 2 commits [8 months ago](#)

 [\[Bugfix\] Migrate to REGEX Library to prevent catastrophic backtracking](#) ... [3f77bcc](#)

 Merge remote-tracking branch
'upstream/main' ✖ [2da2ee5](#)

  **Crucifixion-Fxl** force-pushed the **main** branch from
[619ea88](#) to [2da2ee5](#) [8 months ago](#) [Compare](#)

 **Crucifixion-Fxl** added 2 commits [8 months ago](#)

 [\[Bugfix\] Migrate to REGEX Library to prevent catastrophic backtracking](#) ... ✖ [9523077](#)

 Merge remote-tracking branch
'upstream/main' ✖ [c79c7fe](#)



mergify bot commented on May 23, 2025

This pull request has merge conflicts that must be resolved before it can be merged. Please rebase the PR, [@Crucifixion-Fxl](#).

<https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/working-with-forks/syncing-a-fork>

  **mergify** (bot) added the **needs-rebase** label on May 23, 2025

  Merge remote-tracking branch 'upstream/main'  [24b74b0](#)

  **mergify** (bot) removed the **needs-rebase** label on May 23, 2025



 **mgoin** approved these changes on May 24, 2025 [View reviewed changes](#)

mgoin left a comment

Member

Great! LGTM with green CI.

In the future we could maybe introduce a check to the pre-commit to ensure no `import re` gets added and the user is instructed to use the regex package

 1

  **mgoin** added the **ready** label on May 24, 2025

  **simon-mo** merged commit **4fc1bf8** into [vllm-project:main](#) on May 24, 2025 [View details](#)
99 of 101 checks passed

  **github-project-automation** (bot) moved this to **Done** in **Tool Calling** on May 24, 2025

  **github-project-automation** (bot) moved this to **Done** in **Structured Output** on May 24, 2025

 **zzzyq** pushed a commit to [zzzyq/vllm](#) that referenced this pull request on May 24, 2025

 [\[Bugfix\] Migrate to REGEX Library to prevent catastrophic backtracking \(](#) [bef3250](#)

...

  **aarnphm** mentioned this pull request on May 24, 2025

[CI] enforce import regex instead of re #18665

Merged

gshtras added a commit to ROCm/vllm that referenced this pull request on May 28, 2025



Upstream merge 2025

Partially verified

1900335

05 27 (#557) ...



kexinoh commented on May 28, 2025

Hello! I'm the discoverer of this vulnerability ([GHSA-w6q7-j642-7c25](#)). Thank you very much for your contribution.

However, during my tests on the regex, I found that the current time complexity is $O(N^2)$.

Repeat Count	Length	Time (s)	Match Result
250	2005	0.0560	False
500	4005	0.2325	False
1000	8005	0.9879	False
2000	16005	3.9892	False

If possible, you could try to optimize it to $O(N)$, which would significantly enhance the speed of vllm. Common approaches may include splitting it into multiple regular expressions or exploring other methods? @Crucifixion-Fxl

dtrifiro mentioned this pull request on May 30, 2025

[CI/Build] fix ROCm build #18944

Closed



dtrifiro commented on May 30, 2025

Contributor

This PR broke the ROCm build (and possibly others) due to how build dependencies are handled differently for different target devices.

ROCm fix is here: [#18944](#)

dtrifiro mentioned this pull request on May 30, 2025

[CI/Build] remove regex from build dependencies #18945

Merged



dtrifiro reviewed [on May 30, 2025](#) [View reviewed changes](#)

setup.py

11	10	from pathlib import Path
12	11	from shutil import which
13	12	
	13	+ import regex as re

dtrifiro [on May 30, 2025](#)

Contributor

Can we revert this? This is painful to have and definitely not needed or a security vulnerability.

Revert PR: [#18945](#)

1



wseaton mentioned this pull request [on May 30, 2025](#)

Tool parser regex timeout handling [#18960](#) [Merged](#)



mfindra commented [on Jun 3, 2025](#)

Hey, will there be a CVE assigned for this vulnerability?
thanks



chaunceyjiang mentioned this pull request [on Jul 10, 2025](#)

[Model] Add reason parser for Hunyuan A13B Model. [#20625](#) [Merged](#)



renovate bot mentioned this pull request [on Aug 17, 2025](#)

Update dependency vllm to v0.14.1 [Open](#)
[SECURITY] redhat-ai-dev/developer-images#45



renovate bot mentioned this pull request [on Sep 9, 2025](#)

Update dependency vllm to v0.14.1 [Open](#)
[SECURITY] Jdubrick/ai-developer-images#43